TASK 1

READ TEMPERATURE FROM A SENSOR AND DISPLAY CURRENT VALUE ON LCD

Summary

This circuit is designed to measure temperature using an LM35 temperature sensor and display the readings on a 16x2 I2C LCD. The circuit is controlled by an Arduino UNO, which reads the analog temperature data from the sensor, processes it, and outputs the temperature in Celsius to both the serial monitor and the LCD display.

Component List

1. **Arduino UNO**
   - **Description**: A microcontroller board based on the ATmega328P. It has 14 digital input/output pins, 6 analog inputs, a 16 MHz quartz crystal, a USB connection, a power jack, an ICSP header, and a reset button.
   - **Purpose**: Acts as the main controller for the circuit, reading sensor data and controlling the LCD display.
2. **16x2 I2C LCD**
   - **Description**: A 16x2 character LCD display with I2C interface, which reduces the number of pins required for connection.
   - **Purpose**: Displays the temperature readings obtained from the LM35 sensor.
3. **Temperature Sensor (LM35)**
   - **Description**: A precision integrated-circuit temperature sensor, whose output voltage is linearly proportional to the Celsius temperature.
   - **Purpose**: Measures the ambient temperature and provides an analog voltage output corresponding to the temperature.

Wiring Details

Arduino UNO

- **GND**: Connected to the GND pins of both the 16x2 I2C LCD and the LM35 temperature sensor.
- **5V**: Supplies power to the VCC pin of the 16x2 I2C LCD and the +Vs pin of the LM35 temperature sensor.

- **A0**: Receives the analog output from the Vout pin of the LM35 temperature sensor.
- **A4 (SDA)**: Connected to the SDA pin of the 16x2 I2C LCD for data transmission.
- **A5 (SCL)**: Connected to the SCL pin of the 16x2 I2C LCD for clock signal.

16x2 I2C LCD

- **GND**: Connected to the GND pin of the Arduino UNO.
- **VCC**: Connected to the 5V pin of the Arduino UNO.
- **SDA**: Connected to the A4 pin of the Arduino UNO.
- **SCL**: Connected to the A5 pin of the Arduino UNO.

Temperature Sensor (LM35)

- **+Vs**: Connected to the 5V pin of the Arduino UNO.
- **Vout**: Connected to the A0 pin of the Arduino UNO.
- **GND**: Connected to the GND pin of the Arduino UNO.

Code Documentation

```
#include <Wire.h>
#include <LiquidCrystal_I2C.h>

// Initialize the LCD with I2C address 0x27, 16 columns, and 2 rows LiquidCrystal_I2C lcd(0x27, 16, 2);

void setup() {
    // Start serial communication
    Serial.begin(9600);

    // Initialize LCD lcd.init();
    lcd.backlight();

    // Print message lcd.setCursor(0,0);
    lcd.print("Temp: ");
}

void loop() {
    // Read LM35 sensor value int val =
    analogRead(A0);

    // Convert to voltage
    float voltage = val * (5.0 / 1023.0);

    // Convert voltage to temperature float tempC = voltage
    * 100;

    // Serial Monitor output
    Serial.print("Temperature: ");
    Serial.print(tempC); Serial.println(" C");
```

```
    // LCD output lcd.setCursor(6,0);
    lcd.print(tempC); lcd.print(" C   ");

    delay(1000);
}
```

Code Explanation

- **Libraries**: The code includes the Wire.h library for I2C communication and the LiquidCrystal_I2C.h library for controlling the LCD.
- **LCD Initialization**: The LCD is initialized with the I2C address 0x27, and the backlight is turned on.
- **Setup Function**: Serial communication is started at 9600 baud rate, and the initial message "Temp: " is printed on the LCD.
- **Loop Function**:
  - The analog value from the LM35 sensor is read from pin A0.
  - This value is converted to a voltage, and then to a temperature in Celsius.
  - The temperature is printed to the serial monitor and updated on the LCD every second.